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Soundless: A study on noise sleep disturbance

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Introduction

Sleep quality is a crucial factor in maintaining overall health. In todays world, where a significant portion of the population resides in densely populated urban areas (megalopolises), people are continuously exposed to noise pollution. This constant exposure to ambient noiseday and nightraises concerns about its potential impact on sleep. Investigating how environmental noise affects sleep quality is therefore a highly relevant topic. In this document, we present a study focused on this issue, primarily through the analysis of existing datasets.

We begin by working with the Human Sleep Project dataset [2], developed by Massachusetts General Hospital, which includes around 25,000 electroencephalogram (EEG) recordings from subjects with various sleep pathologies. These recordings, stored in the .edf (European Data Format), often come with detailed annotations and information on the EEG channels used. The dataset has been widely studied by researchers across multiple disciplines. Leveraging this resource, we aim to develop an artificial intelligence-based solution capable of automatically annotating EEG recordings to assist in sleep quality analysis.

Next, we introduce the Soundless [1] dataset, part of a research project led by Universitat Rovira i Virgili in Tarragona, Spain. This project is designed to study the impact of environmental noise on sleep quality through the use of FitBit devices, which monitor participants' vital signs during sleep, and a companion Android application that records ambient sound throughout the night. While the dataset currently includes data from approximately 40 participants, it provides valuable real-world insight into how different noise conditions may influence various stages of sleep. It also acknowledges that individuals may have different sensitivities to noise, and that certain noise frequencies might affect sleep differently across subjects.

The objective of this work is to bridge the controlled, large-scale data of the Human Sleep Project with the real-world, noise-focused observations of the Soundless dataset. By connecting these two sources, we aim to extrapolate key metrics and draw meaningful conclusions about the relationship between environmental noise and sleep quality.

The following pages present an initial approach to studying sleep quality by combining data-driven analysis with real-world noise exposure scenarios. The primary aim of this work is to explore how different stages of sleep are influenced by environmental noise, using a combination of two complementary datasets: the Human Sleep Project and the Soundless dataset. Through this exploratory study, we aim to build a foundation for further research into sleep quality assessment in urban and noise-rich environments.

The main goals of this document are as follows:

- Design a method for analysing sleep stages in subjects from the Human Sleep Project dataset. This includes processing and interpreting electroencephalogram (EEG) signals to identify key sleep phases such as REM, light sleep, and deep sleep, with a focus on creating a robust and scalable analysis pipeline.
- Define a set of metrics derived from the identified sleep stages to establish a meaningful connection between participants in the Soundless dataset and those in the Human Sleep Project. These metrics may include sleep stage duration, frequency of transitions between stages, interruptions, and overall sleep efficiency, among others.
- Draw data-driven conclusions about the quality of sleep in environments with varying levels of noise. By correlating ambient sound data with changes in sleep patterns, we aim to understand how specific noise characteristics (e.g., intensity, frequency, duration) may affect different individuals and sleep stages, ultimately influencing overall sleep quality.

This preliminary study sets the groundwork for future research that could lead to better tools for monitoring and improving sleep in urban settings, potentially informing both clinical practice and consumer health applications.

State of art

The relationship between environmental noise and sleep quality has been the focus of extensive interdisciplinary research due to its profound implications for public health. Numerous studies have demonstrated that nocturnal noise exposure adversely affects both the physiological and psychological dimensions of sleep, including sleep architecture, duration, and subjective sleep quality.

One of the most influential sources on this topic is the World Health Organization's environmental noise guidelines, which synthesize extensive epidemiological and experimental evidence. Basner and McGuire (2018), in their systematic review, concluded that night time exposure to noise from traffic, aircraft, and industrial sources leads to delayed sleep onset, increased number and duration of awakenings, and reduced deep sleep (slow-wave sleep). These findings support earlier work by Muzet (2007), who outlined the biological pathways through which noise interferes with sleep, such as through increased heart rate, blood pressure, and cortical arousals.

Different types of environmental noise have been shown to affect sleep in varying degrees. Griefahn et al. (2006) and Pirrera, De Valck, and Cluydts (2010) reviewed the impacts of road, rail, and air traffic noise, noting that aircraft noise tends to be the most disruptive due to its variability and unpredictability. Basner et al. (2008) used polysomnography to demonstrate that aircraft noise exposure reduces the proportion of REM and slow-wave sleep, both critical for physical recovery and cognitive function.

Recent studies have also investigated the role of individual differences in noise sensitivity. Elmenhorst et al. (2012) found that individuals with higher subjective sensitivity to noise are more prone to awakenings and have lower sleep efficiency under noisy conditions. Similarly, Riedel et al. (2022) showed that noise annoyance significantly moderates the relationship between traffic noise exposure and sleep quality, indicating that psychological factors mediate the perception and physiological impact of noise.

Urbanization and socio-economic status are additional contextual factors that modulate the effects of noise. Dzhambov et al. (2017), through meta-analysis, found that people living in urban and low-income neighbourhoods are disproportionately affected by road traffic noise, suggesting environmental justice implications. Brown and van Kamp (2009) emphasized the role of subjective appraisal and context in shaping noise response, indicating that perceptions of control and expectations may alter physiological responses to the same noise level.

The long-term consequences of chronic noise-induced sleep disruption are also well-documented. Witte et al. (2021) linked prolonged night-time noise exposure to cognitive decline in older adults, while Dratva et al. (2012) showed that noise-related sleep disturbance is associated with reduced health-related quality of life and increased psychological distress.

In sum, the current body of research provides robust evidence that environmental noise impairs sleep quality through both direct physiological mechanisms and mediated psychological pathways. While regulation and mitigation strategies have been proposed, further research is needed to explore personalized interventions and to better understand vulnerable populations.

To the best of our knowledge, no prior research has established a relationship between the Human Sleep Project and the Soundless dataset. This study introduces a novel methodological approach aimed at bridging these two datasets, with the objective of advancing our understanding of how environmental noise influences sleep quality.

Analisis

A critical step in this study involves understanding and preprocessing the data from the Human Sleep Project. Given the scale and complexity of this dataset, careful examination is required before any meaningful analysis can be performed.

EEG montages are organized arrangements of channels, where each channel represents the potential difference between two electrodes, used to display and interpret scalp electrical activity. They are crucial for visualizing brain activity, enabling the appreciation of waveforms from different perspectives, aiding in lateralization and localization of potential sources, and facilitating artifact identification.

To study the Human Sleep Project dataset, the first task is to analyze its vast and highly heterogeneous structure. The dataset consists of over 25,000 electroencephalogram (EEG) recordings, each captured using different channel configurations. As a starting point, it is essential to understand the various EEG montages used across recordings and to develop a unified framework that allows consistent analysis across the dataset. Within the collection, there are 23 distinct montage configurations, each comprising the same number of channels but often differing in the specific channels included. To address this variability, I conducted an analysis of channel frequency across all configurations, aiming to identify a common subset that would enable a standardized and comprehensive study of the data.

The study yields the following results:

Channel	Frequency (%)
ABD	88.719512
AIRFLOW	75.304878
C3-M2	71.951220
CHEST	87.042683
E1-M2	84.451220
EKG	87.957317
HR	89.786585
IC	89.329268
LAT	82.164634
O1-M2	70.731707
RAT	79.878049
SNORE	91.920732
SaO2	90.396341

Table 1: Channel frequencies

In table 1 we can see the most common channels in the dataset, and the associated frequency of the channel across all the electroencephalograms in the dataset. With this information we can impose that our electroencephalograms have, at least, this channels and we must get a representation of around 71% of the dataset.

Going one step further we can impose channel equivalences. We define a channel equivalence as a pair of channels that, if exchanged between configurations, we obtain the same configuration. Studying the different configurations present in the dataset we obtain the following equivalences:

Channel equivalences								
C4-M1 C4-M2	CHIN1-CHIN2 CHIN1-CHIN3 CHIN2-CHIN3 CHIN3-CHIN1 CHIN3-CHIN2 Chin1-31 Chin1-Chin2 Chin1-Chin3 Chin1-P3 Chin1-P4 Chin2-Chin3 Chin3-Chin2	E1-M1 E1-M2	E2-M1 E2-M2	EKG EKG-E1	F4-M1 F4-M2 F8-M1 Fp2-M1 Fp2-M2	O2-M2 O1-M1 O2-M1	LAT LAT-E1	RAT RAT-E1

Channel	Frequency (%)	Equivalence
ABD	88.719512	No
AIRFLOW	75.304878	No
C3-M2	71.951220	No
C4-M1	76.371951	Yes
CHEST	87.042683	No
CHIN1-CHIN2	91.310976	Yes
E1-M1	85.365854	Yes
E2-M1	87.347561	Yes
EKG	90.396341	Yes
F4-M1	81.402439	Yes
HR	89.786585	No
IC	89.329268	No
LAT	84.451220	Yes
O1-M2	70.731707	No
O2-M2	86.280488	Yes
RAT	82.164634	Yes
SNORE	91.920732	No
SaO2	90.396341	No

Table 2: Frequencies with equivalences aggregated

In table Table 2 I summarize the frequencies obtained when aggregating the equivalences found by the code, after all we have 18 meaningful channels to study, however we need to take into account that most of them are not related to brain activity, so looking for brain activity on those channels will make no sense. However we can try to use the 18 channels to feed a neural network as they are, as a sequence of voltages per channel over time.

Power Spectral Density (PSD) analysis is a widely used quantitative EEG (qEEG) technique for examining the spectral composition of EEG signals during sleep. It provides valuable insights into various sleep stages and can aid in the identification of potential sleep disorders by measuring the distribution of power across different frequency bands within the EEG signal.

Delta activity

Delta brainwaves, also known as delta activity, are the slowest brainwaves, typically between 0.5 and 4.5 Hz. They are primarily associated with deep, dreamless sleep, where the body is in a state of complete relaxation and the mind is minimally active. Delta waves are also observed in newborns, and can be induced in a waking state by experienced meditators.

They usually show a high amplitude and are most prevalent during deep sleep stages (3), where the brain is essentially shut down for repair and regeneration. Delta waves are linked to various restorative processes, including tissue repair, muscle growth, and hormone release.They are also involved in transferring information from the hippo campus to the neocortex, aiding in long-term memory storage.

Theta activity

Theta brainwaves, characterized by frequencies between 4.5-8.5 Hz, are associated with states of relaxation, deep meditation, and creativity. They are often observed during sleep, particularly during dream sleep, and are also linked to states of daydreaming and focused attention. Theta activity is thought to play a role in processing information, making memories, and facilitating intuitive insights.

Disrupted or divergent theta activity has been linked to various mental health conditions.

Alpha activity

Alpha brainwaves, with a frequency of 8.5-11.5 Hz, are associated with a relaxed, awake state, often observed during activities like daydreaming or meditation. They indicate the brain is active but not overly focused, promoting feelings of calmness and ease.

Alpha activity is typically strongest over the posterior areas of the head, particularly the occipital cortex. Alpha waves can be influenced by various factors, including sensory stimuli, mental activities, and even the position of the eyes (they are often more prominent with eyes closed).

Sigma activity

Sigma activity, also known as sigma power or spindle activity, refers to brainwave activity in the frequency range of 11.5-15.5 Hz, a key component of non-REM sleep. These brainwaves are characterized by transient oscillations and are often associated with sleep-dependent memory consolidation and sleep stability. In infants, sigma activity is linked to psycho motor development and local brain maturation.

Beta activity

Beta activity in the brain refers to a range of brainwave frequencies, typically between 15.5 and 30 Hz, associated with alert, focused, and active mental states. These brainwaves are commonly present during waking hours and when engaged in tasks requiring concentration, problem-solving, or decision-making.

An excessive beta activity can be associated with symptoms of brain over-arousal, such as anxiety, obsessiveness, and sleep difficulties, while a deficient beta activity may be linked to difficulties concentrating, problem-solving, and overall brain under-arousal.

Motivated by findings from recent literature [3], we can apply PSD analysis to extract the frequency spectrum from the EEG recordings, resulting in a feature vector that quantifies the power observed in each channel across frequency bands associated with brain activity. However, upon closer examination, it became evident that the set of EEG channels initially considered was not optimal for detecting brain-related spectral features. Therefore, the list of channels used in the analysis was refined and reduced to a smaller, more representative subset. The selected channels used for PSD analysis are as follows:

Channel	Frequency (%)	Equivalence
C3-M2	71.951220	No
C4-M1	76.371951	Yes
E1-M1	85.365854	Yes
E2-M1	87.347561	Yes
F4-M1	81.402439	Yes
O1-M2	70.731707	No
O2-M2	86.280488	Yes

Table 3: Channels for PSD Analisys

Design

In order to establish a meaningful relationship between the Human Sleep Project dataset and the Soundless dataset, it is essential to standardize the analysis of the Human Sleep Project data. This can be achieved by predicting the sleep stage of a subject at each moment within their electroencephalogram (EEG) recording. Once sleep stages are accurately classified, relevant sleep metrics can be computed, enabling comparisons between the two datasets.

To perform this classification, a neural network presents itself as a suitable tool due to its capacity to model complex patterns in time-series data. The goal is to classify each segment of the EEG into one of the following sleep stages:

- **Sleep_stage_W:** This stage indicates that the subject is awake.
- **Sleep_stage_N1:** This stage indicates that the subject is in the first sleep stage.
- **Sleep_stage_N2:** This stage indicates that the subject is in the second sleep stage.
- **Sleep_stage_N3:** This stage indicates that the subject is in the third sleep stage.
- **Sleep_stage_R:** This stage indicates that the subject is in REM (Rapid Eye Movement) stage.

As stated in previous sections, the initial approach adopted was to analyze the electroencephalograms in their original form, treating them as sequences of voltage measurements recorded over time for each channel. To this end, I employed a fully convolutional neural network (FCN), which is particularly well-suited for modeling temporal dynamics and spatial relationships present within multi-channel EEG data.

The choice of an FCN was motivated by its ability to efficiently extract features from time-series data without relying on traditional recurrent architectures. This approach leverages convolutional filters to capture both short-term fluctuations and longer-term dependencies within the signal, thereby enabling a comprehensive analysis of the electroencephalogram's intrinsic structure. Furthermore, FCNs have been demonstrated to perform robustly in similar biomedical applications, making them an attractive option for tasks such as sleep stage classification and signal anomaly detection.

By processing the EEG recordings in their raw format, this method allows for an end-to-end feature extraction and classification framework, whereby the network learns to identify subtle patterns associated with different sleep stages directly from the voltage sequences. The following sections detail the network architecture, the training regimen, and the evaluation metrics used to assess the performance of this approach in analyzing and classifying the EEG data.

Fully Convolutional neural network (FCN)

Based on the paper Time Series Classification from Scratch with Deep Neural Networks: A Strong Baseline [4] and the related work found at [5] I build a fully convolutional network for time series classification, the architecture is as follows:

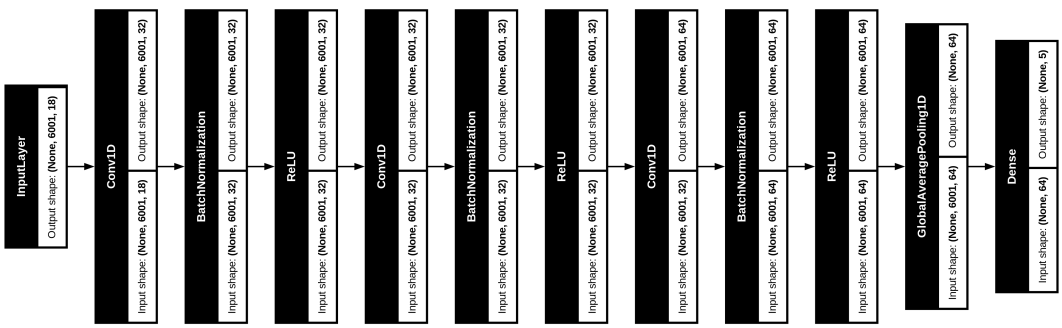

Figure 1: FCN architecture

The input shape to the fully convolutional neural network is (6001, 18), where 18 corresponds to the number of EEG channels and 6001 to the number of time points in each segment. Each input segment represents a 30-second window of the EEG signal, sampled at 200 Hz, resulting in 6000 samples, plus one additional point depending on the indexing used during preprocessing. For consistency, only EEG recordings sampled at 200 Hz are retained; any files with a different sampling rate are excluded from this approach.

The network architecture begins with a 1D convolutional layer consisting of 32 filters and a kernel size of 3, applied along the temporal axis. This is followed by a second convolutional layer, also with 32 filters, but using a kernel size of 5 to capture slightly broader temporal features. A third convolutional layer with 64 filters and a kernel size of 5 is then used to further increase the representational capacity of the network.

To reduce dimensionality and extract global temporal features, an average pooling layer is applied after the convolutional blocks. Finally, the output is passed through a dense (fully connected) layer with 5 neurons, corresponding to the number of target sleep stages or classes in the classification task.

The activation function used for all hidden layers in the network is the Rectified Linear Unit (ReLU), chosen for its computational efficiency and its ability to mitigate the vanishing gradient problem during training. For the output layer, a softmax activation function is applied, as the task involves multi-class classification, with the network output representing a probability distribution over the defined sleep stage classes.

The hyperparameters of the neural network were optimized using Keras Tuner, a tool designed to automate the selection of optimal configurations for deep learning models. While Keras Tuner provides a structured and efficient way to explore hyperparameter spaces, its integration with this project posed certain limitations. Specifically, because the full dataset is large and hosted externally, it was not feasible to perform tuning across the entire dataset. As a result, the tuning process was conducted on a representative subset of the data. Despite this constraint, Keras Tuner proved to be a valuable aid in guiding the selection of key model parameters such as the number of filters, kernel sizes, and learning rates.

Several optimization strategies were evaluated to determine the most effective training configuration for the proposed network architecture. In particular, the Adam, Stochastic Gradient Descent (SGD), and AdamW optimizers were tested. For the AdamW optimizer, a weight decay parameter of $1 * 10^{-4}$ was applied to regularize the model and prevent overfitting.

Additionally, a range of learning rates between $1 * 10^{-3}$ and $1 * 10^{-5}$ was explored during training to assess their impact on convergence and performance. A learning rate scheduler was also employed to dynamically adjust the learning rate over time, further improving training stability and efficiency.

LSTM

Long Short-Term Memory (LSTM) networks [6] are a type of recurrent neural network (RNN) specifically designed to model and learn from sequential data. Unlike traditional RNNs, which suffer from vanishing or exploding gradients over long sequences, LSTMs incorporate memory cells and gating mechanisms that allow them to capture long-range dependencies and temporal dynamics more effectively. This makes them particularly well-suited for time series classification tasks, such as analyzing electroencephalogram (EEG) signals, where temporal patterns play a crucial role in identifying different sleep stages.

Given these strengths, an alternative model architecture was developed by replacing the final two convolutional layers in the original network with one or more LSTM layers. The rationale behind this change is that while convolutional layers are effective at extracting local spatial and short-term temporal features, LSTMs may provide improved performance by explicitly modeling the temporal evolution of brain activity across the EEG signal. This hybrid approach leverages the strength of convolutional layers for feature extraction at the lower levels of the network, while utilizing LSTMs to capture higher-order temporal dependencies at later stages.

The architecture proposed is as follows:

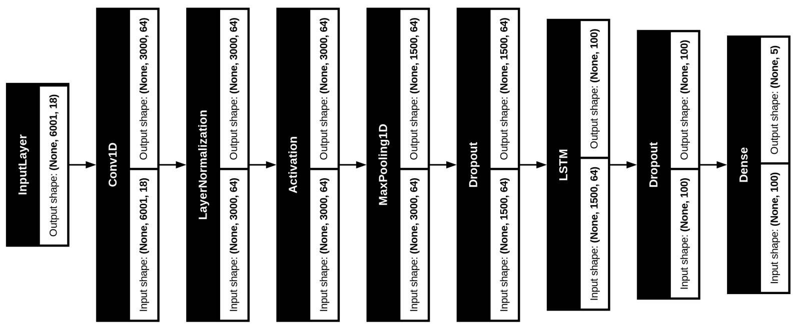

Figure 2: CN-LSTM architecture

The input layer remains unchanged, with a shape of (6001, 18), corresponding to 30-second EEG segments sampled at 200 Hz across 18 channels. The network begins with an initial 1D convolutional layer consisting of 64 filters and a kernel size of 3, which is effective for capturing local temporal patterns in the data. This is followed by a normalization layer, which helps stabilize the learning process, and a ReLU activation function to introduce non-linearity. A pooling layer is then applied to downsample the feature maps and reduce computational complexity.

To mitigate overfitting, a dropout layer is added next. This layer randomly deactivates a fraction of the neurons during training, helping to prevent the model from becoming too sensitive to noise in the input data and promoting generalization. Following this, a Long Short-Term Memory (LSTM) layer with 100 units is introduced, using the default tanh activation function, to capture the sequential dependencies in the EEG signal.

Another dropout layer is applied after the LSTM to further regularize the model. Finally, the architecture concludes with a dense output layer containing 5 neurons, corresponding to the number of sleep stage classes in the classification task. This output layer uses a softmax activation function to produce class probabilities.

For this architecture, several optimizers were evaluated, including Adam with learning rates ranging from $1 * 10^{-3}$ to $1 * 10^{-5}$, as well as AdamW, incorporating a weight decay parameter of $1 * 10^{-4}$ to improve regularization.

Multi Layer Perceptron (MLP)

A Multi-Layer Perceptron (MLP) [7] is a type of feedforward neural network characterized by an input layer, one or more hidden layers, and an output layer. MLPs are particularly well-suited for classification tasks involving tabular or fixed-length feature data.

Following the Power Spectral Density (PSD) analysis, we obtain a fixed-length feature vector representing the power distribution across frequency bands related to brain activity for each selected EEG channel. Given this structured feature representation, a straightforward MLP architecture is an appropriate choice for classification. The specific architecture employed in this study is as follows:

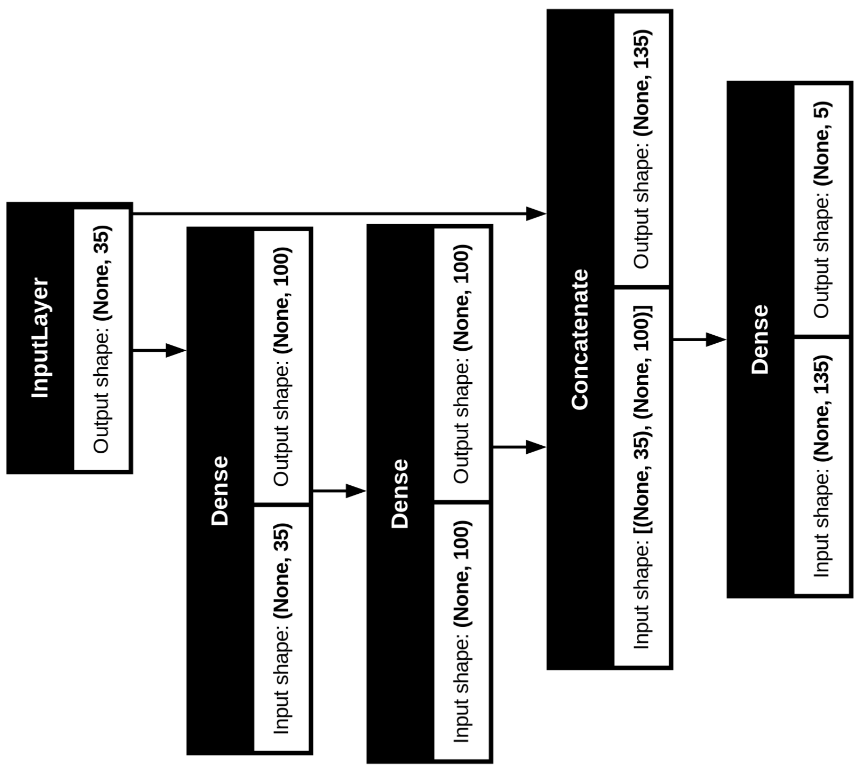

Figure 3: MLP architecture

The input layer has a shape of (35), corresponding to the 7 selected EEG channels multiplied by the 5 frequency bands of interest derived from the PSD analysis. This input is first passed through a dense layer with 100 neurons and a ReLU activation function, followed by a second hidden dense layer, also with 100 neurons and ReLU activation.

After these hidden layers, the original input vector is concatenated with the output of the second hidden layer, creating a model with both a deep path (through the hidden layers) and a wide path (direct input). This architecture leverages both the learned high-level features and the raw input features, which can enhance classification performance when the input features are strongly representative of the target classes.

Finally, the concatenated output is passed to a dense output layer with 5 neurons and a softmax activation function, corresponding to the five sleep stage classes to be predicted.

The optimizer selected for training the MLP is Adam, initialized with a learning rate of $1 * 10^{-3}$. To enhance training stability and promote efficient convergence, a ReduceLROnPlateau learning rate scheduler is employed. This scheduler reduces the learning rate automatically when the validation error plateaus, allowing the network to focus more effectively on minimizing the loss and achieve a more stable, albeit slower, convergence.

The hyper parameters, including the number of neurons and the learning rate for this network, were optimized using Keras Tuner. Multi-Layer Perceptrons are highly sensitive to data scaling, as they typically assume input features are centered around zero with a standard deviation of one. To meet this assumption, the data preprocessing pipeline first normalizes the samples to have zero mean and unit variance, followed by scaling each sample using a MinMaxScaler. This two-step scaling process significantly improved the stability of the network during training and led to a notable increase in classification accuracy.

Implementation

This section provides an overview of the technologies and tools used to implement the methodology described above. Given the size and complexity of the dataset, comprising over 25,000 files, each averaging approximately 500 MB, the neural network training process is non-trivial. It involves multiple stages: first, downloading the data; second, preprocessing and aggregating the data; and finally, training the neural network models.

The entire implementation is developed in Python, a widely-used programming language for artificial intelligence and machine learning tasks. Pythons rich ecosystem of libraries, including Keras, TensorFlow, and scikit-learn, are leveraged extensively throughout this project.

For the data acquisition phase, the boto3 library is utilized to interface with AWS S3 storage services, from which the dataset files are downloaded.

Data preprocessing is facilitated by the MNE library, which is specifically designed for handling electrophysiological data formats such as .edf files. MNE enables efficient extraction of EEG epochs corresponding to specific annotations and the computation of Power Spectral Densities (PSDs) for each event.

To manage and organize the processed data, a PostgreSQL relational database is employed. Preprocessed data is aggregated into batches and stored within this database, allowing the neural network training routines to efficiently access and load data in manageable chunks.

Finally, the neural networks themselves are implemented using Keras, a high-level deep learning library that abstracts the complexities of model creation and training, enabling rapid prototyping and experimentation.

Evaluation

At the moment of writing, the MLP has been trained with 1437 subjects, with, in average, around 850 events each patient what results in more that 1.2M events. We are reaching an accuracy of nearly a 70% in the validation phase. Now, the tests have been performed in 124 subjects taken randomly from the dataset, with about 102000 events, those subjects are nod feed into the neural network at training stage.

The classification report obtained for the predictions computed with the resulting model is the following:

	precision	recall	f1-score	support
Sleep_stage_W	0.59	0.60	0.60	13843
Sleep_stage_N1	0.33	0.26	0.29	10978
Sleep_stage_N2	0.80	0.77	0.78	47168
Sleep_stage_N3	0.70	0.73	0.71	13242
Sleep_stage_R	0.65	0.78	0.71	17240
accuracy			0.69	102471
macro avg	0.61	0.63	0.62	102471
weighted avg	0.68	0.69	0.68	102471

Table 4: Classification report

As we can see, our main problem is with class Sleep_stage_N1, where we get a precision of 33%, if we look closely into the classification report we observe that the class is the class with less support, probably applying some data augmentation techniques, like SMOTE will help the neural network to predict better for this class.

The confusion matrix obtained in the training is the following:

Figure 4: Confussion matrix

Here we can appreciate that our classifier is not performing well for the Sleep_stage_N1 class, being confused mainly with the class Sleep_stage_R and Sleep_stage_N2. The rest of the classes shows a better performance, being Sleep_stage_N2, Sleep_stage_N3 and Sleep_stage_R above the 70% of success in the prediction.

Data augmentation

Data augmentation or oversampling is a technique that generates synthetic samples for a given dataset, this synthetic samples are generated according to the samples we already have from the target class. To try to mitigate the problem found in the evaluation section with the class Sleep_stage_N1 I'm generating synthetic samples exclusively for this class to see if the network performance can be improved for this events.

Three approaches have been employed regarding data augmentation, the first one is to generate synthetic samples for the class Sleep_Stage_N1 to equate the number of samples of this class to the number of classes of the majority class, usually Sleep_Stage_N2, the confusion matrix obtained for this approach is the following:

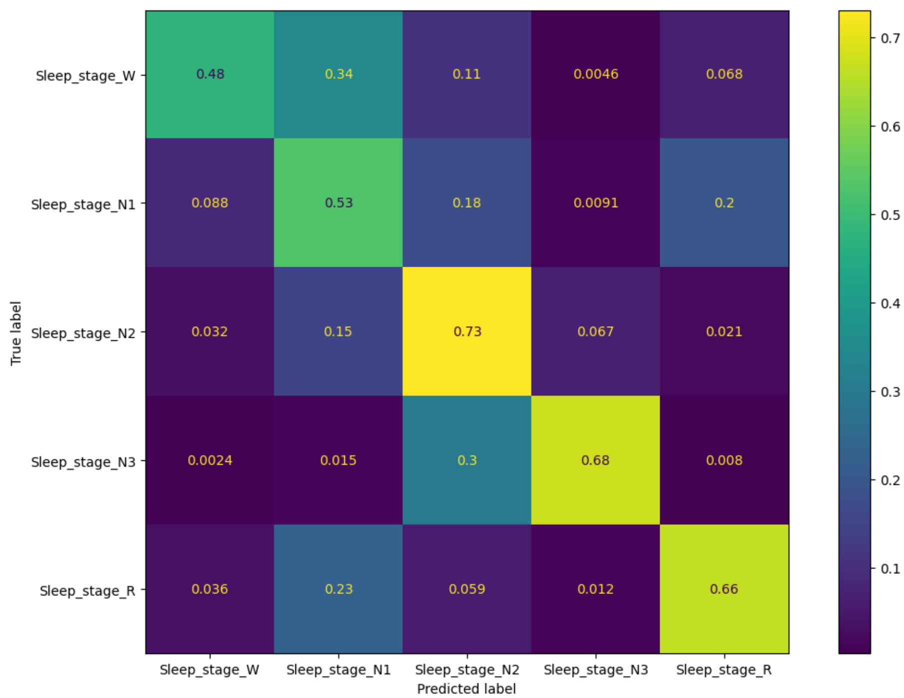

Figure 5: Confusion matrix oversampling N1 to equate the majority class

As we can see in figure 5, we improve the classification for Sleep_stage_N1 to the detriment of other classes, this is no ideal but all the classes but Sleep_stage_W are above the 50%.

In the next approach I computed the mean of the elements for all classes, if Sleep_stage_N1 number of elements was below that number, the class is oversampled to achieve that number, the confusion matrix is the following:

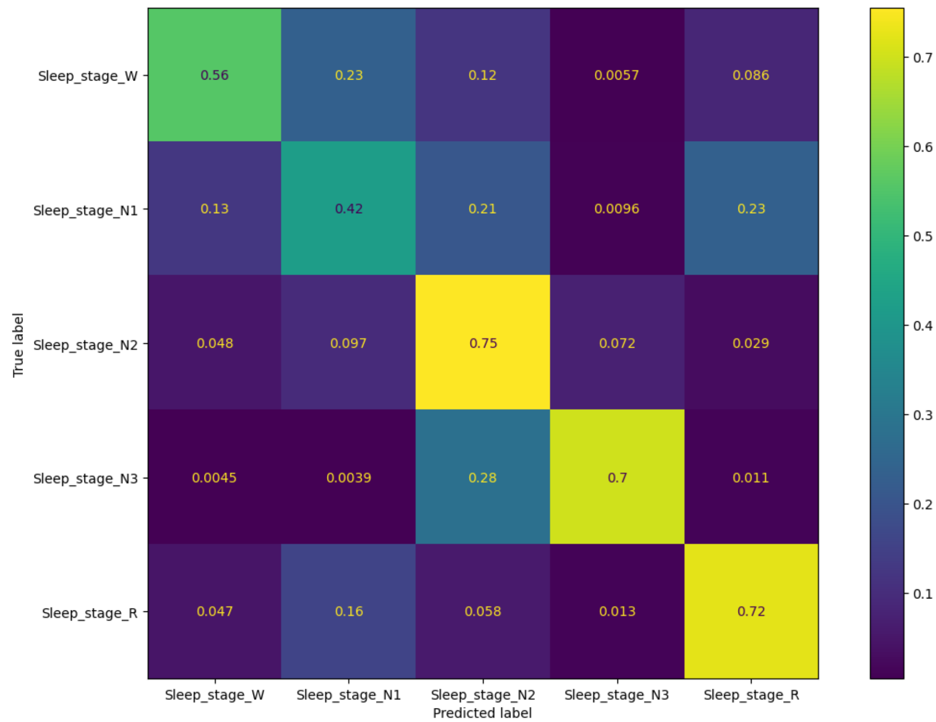

Figure 6: Confusion matrix oversampling N1 to achieve the mean

As we can see in figure 6 this approach is more similar to the one obtained without any oversampling at all, however Sleep_stage_N1 success rate is improved by over a 15%, again in detriment to the other classes. If we oversample just one class, we hurt the rest of the classes, no matter what we do. The third approach that has been tried here is to oversample all the minority classes to match the majority class, it is: the default configuration for SMOTE. The confusion matrix obtained in this case is the following:

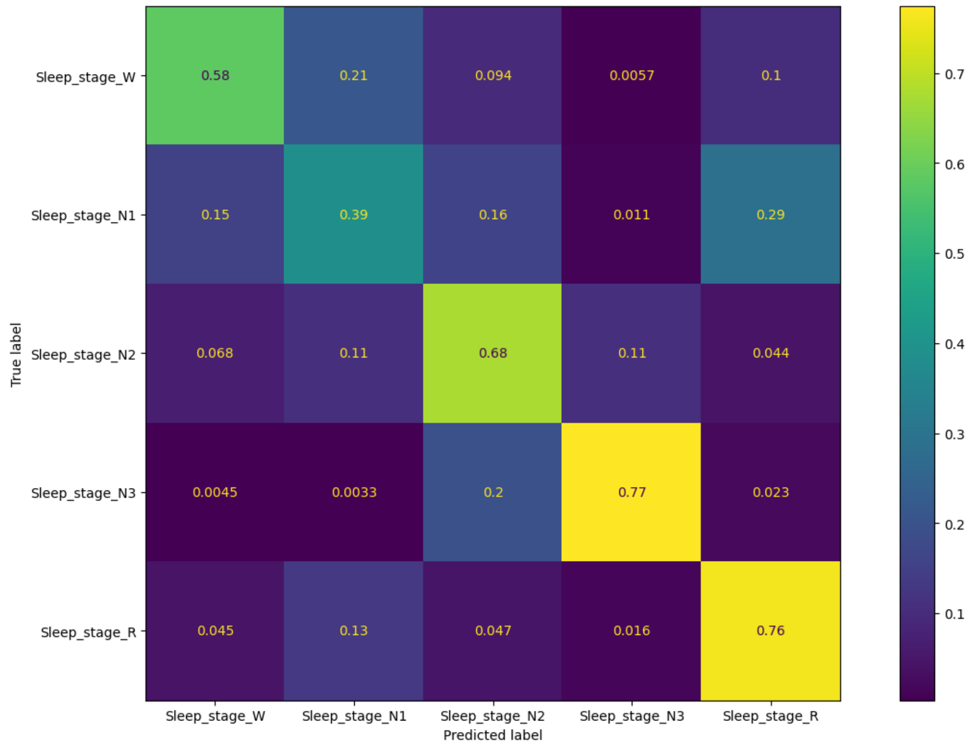

Figure 7: Confussion matrix oversampling all minority classes

As we can see in figure 7, the overall accuracy is reduced, and, in this case we do not improve Sleep_stage_N1 that much, which was the first goal of oversampling at all.

Class clustering

Oversampling did not succeed as expected, another technique we can apply here is to cluster the class Sleep_stage_N1 and Sleep_stage_N2. We loose precision in the results obtained, but we may be more accurate predicting a the stages of sleep. By applying this technique the model was trained on 1371 subjects with a mean of 850 samples per subject, giving a result of 1.16M samples. The same test set used for the previous sections is used here, the classification report obtained in this case is the following:

	precision	recall	f1-score	support
Sleep_stage_W	0.64	0.80	0.72	17240
Sleep_stage_N1/N2	0.83	0.73	0.78	58146
Sleep_stage_N3	0.66	0.76	0.70	13242
Sleep_stage_R	0.58	0.61	0.60	13843
accuracy			0.73	102471
macro avg	0.68	0.73	0.70	102471
weighted avg	0.74	0.73	0.73	102471

Table 5: Classification report

And the confusion matrix obtained is:

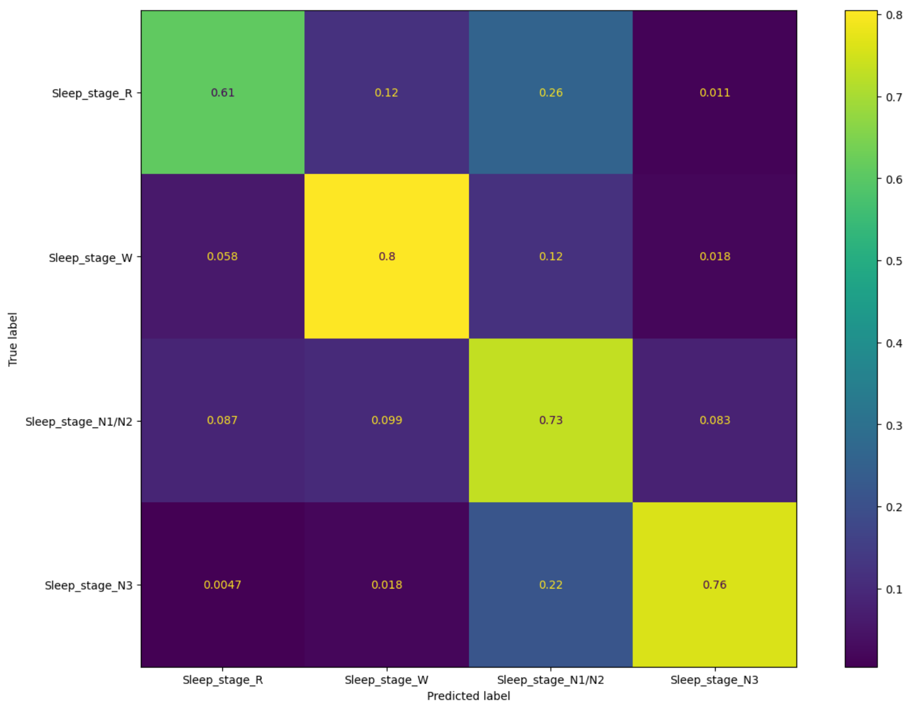

Figure 8: Confusion matrix clustering N1/N2

This time the results are pretty more solid, we will be able to classify the different sleep stages and, particularly we will be able to know when a subject is awake or not. With this kind of data we can carry on and try to link our data set to Soundless dataset.

Soundless

The soundless project carried out by the Universitat Rovira i Virgili (Tarragona, Spain) aims to be a citizen science platform to study how noise affect rest in different neighbourhoods of the city of Tarragona.

The soundless project gathered 40 neighbours of the city of Tarragona living in the neighbourhoods of El Serrallo and Barri del port an conducted a study on how the sleep is disturbed by train lines noise. Tarragona is a port city located in north-east Spain (Catalonia) by the Mediterranean Sea. A major problem is that railway tracks create a barrier between the city and the sea (beach, port). Apartment blocks are located just at 2-5 meters of distance from the tracks without any acoustic barriers.

The first challenge of the Sounless project is to gather the data to conduct an study on sleep quality. To do so they used a FitBit (an intelligent bracelet) able to capture the vital sings of an individual, such as the heart rate, the oxygen saturation, etc. Once the data is gathered the next step is to detect an incidence in a reliable manner, here we define a sleep incident as when a person moves from a deep sleep phase to being completely awake due to excessive noise. To do so several algorithms are applied to the data in order to detect anomalies in sleep related to noise in the ambient. A first approach was to use direct mathematical correlation between hear rate and noise, some factors affected this measurement however:

- Individual Variability: Participants exhibited diverse heart rate patterns and sensitivities to noise.
- Data Complexity: Recordings contained inherent noise even after filtering, adding to the challenge of isolating the influence of sound incidents.
- Reaction Time Differences: Individual response times to noise varied considerably, ranging from 10 to 40 seconds, further complicating correlation analysis.
- Sensitivity Fluctuations: Individual sensitivity to noise appeared to fluctuate, with inconsistent reactions even within the same participant

- Sleep Stage Influences: Natural heart rate variations during different sleep stages added another layer of complexity.

What leaded the soundless team to adopt z-scores algorithm. With this approach they were successfully able to detect incidents in heart rate and in sound measurements, and, afterwards relate the incidents in heart rate to incidents in sound. They results show that the legal limit set by the World Health Organization (WHO) of 45 decibels was exceeded on 98% of nights with recordings. Additionally they show that each and every user have, in average, 12 sound incidents during the night and half of the nights that a recording was taken an incident in their heart rate is detected. Additionally one in five nights each user has been woken up due to a hight level on ambient noise.

Soundless study carry on trying to detect individuals with a higher sensitivity to noise in the ambient, having individuals with values close to 38 dB to have a disturbance and others that requires louder noises. Particularly, they show that 70.59% of the times a sleep disturbance has been detected, it has occurred in people with a higher sensitivity to noise, in this case, a sensitivity less than 45 decibels. On the other hand, people with lower sensitivity to this type of situation have only registered 29.41% of the sleep disturbances caused by noise. After the study on individual persons they examine general sensitivity to noise in the ambient, where they find that values between 40 and 50 dB are able to create incidences in the heart rate and in the sleep process of almost any individual.

The study concludes that the night noise produced by freight trains in the seaside of Tarragona is far beyond legal limits (>45dB) and up to 80dB during all nights. Furthermore, they consistently detected the impact of noise in health variables like heart rates and sleep modes, which affected most users in the study during all nights. This disturbances during sleep are causing irregular heart rates, sudden awakenings, and noticeable impacts in sleep quality in the citizens.

With Soundless data, we can try to compare the users in HSP project and in Soundless dataset to see if noise is affecting the sleep quality of the subjects.

Sleep Metrics

When looking for sensitivity to external stimuli is often good to stick to some well known metrics. The main metric is sleep efficiency, defined as:

$$SE = TST/TIB$$

Where TST is the total time sleeping in bed, and TIB is the total time in bed. Now that we can classify our electroencephalograms slices into the different sleep stages, we can compute the sleep efficiency of a subject, with such metric we will be able to compare soundless users with subjects which recordings have been taken in a very controlled environment, allowing us to elucidate if a recording has been altered by external noise given a user profile.

Taking into account the different sleep phases of the different subjects we can try to tune up even more our study. Deep sleep, also known as slow-wave sleep or stage N3, is a crucial phase of the sleep cycle characterized by its restorative properties. It's the deepest stage of non-REM sleep, where the body and mind engage in restorative processes. During deep sleep, brain waves slow down, and the body experiences a significant decrease in heart rate, breathing rate, and muscle tension. This stage is essential for physical recovery, immune system strengthening, memory consolidation, and hormone regulation. REM (Rapid Eye Movement) sleep is the final stage of a sleep cycle, characterized by intense brain activity, vivid dreams, and temporary paralysis of the limbs. It typically begins about 90 minutes after falling asleep and recurs in cycles throughout the night, with later cycles becoming longer. REM sleep is believed to be crucial for cognitive functions like memory consolidation, learning, and emotional processing.

In addition to the sleep efficiency we will compute the percentage of the night a subject is in light sleep stage (stages N1 and N2), deep sleep stage (stage N3) or REM sleep stage. With this values we can try to aggregate all the users from soundless dataset and The Human Sleep Project and estimate clusters of users. We will work with around 12m sessions of The Human Sleep project and the 362 sessions of soundless dataset.

We will use the well known KMeans algorithm for the clustering. Before clustering the subjects a PCA analysis to reduce the variable space is performed, using two components for the four variables we are studying, the sleep efficiency, the percentage of time in light sleep, the percentage of time in deep sleep and the percentage of time in REM sleep. Further on, performing a silhouette score study on the data for 2, 3, 4, 5 and 6 clusters we obtain that three clusters is the optimum with an score of 0.392, the graphic is as follows:

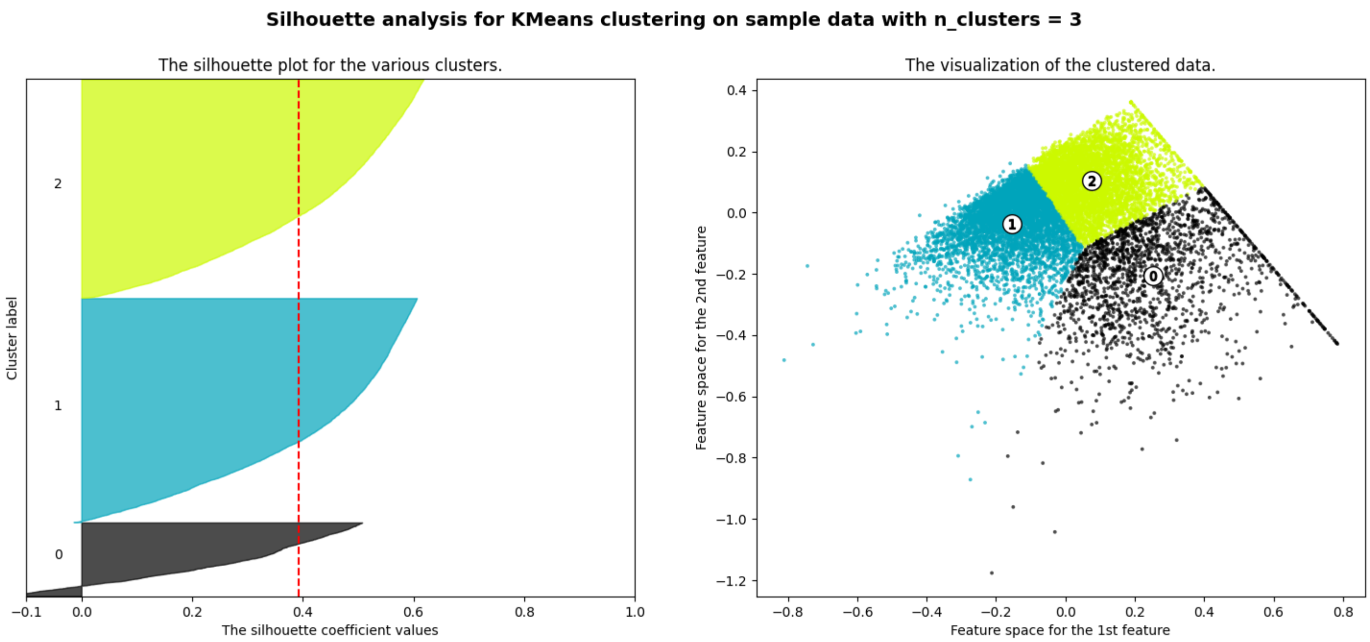

Figure 9: Silhouette score for 3 clusters

The graphics for 2, 4, 5 and 6 clusters are not included in this document for the sake of brevity.

With this information we can carry on our study, performing a KMeans clustering for three clusters we can study the sleep efficiency en each cluster, the results are the following:

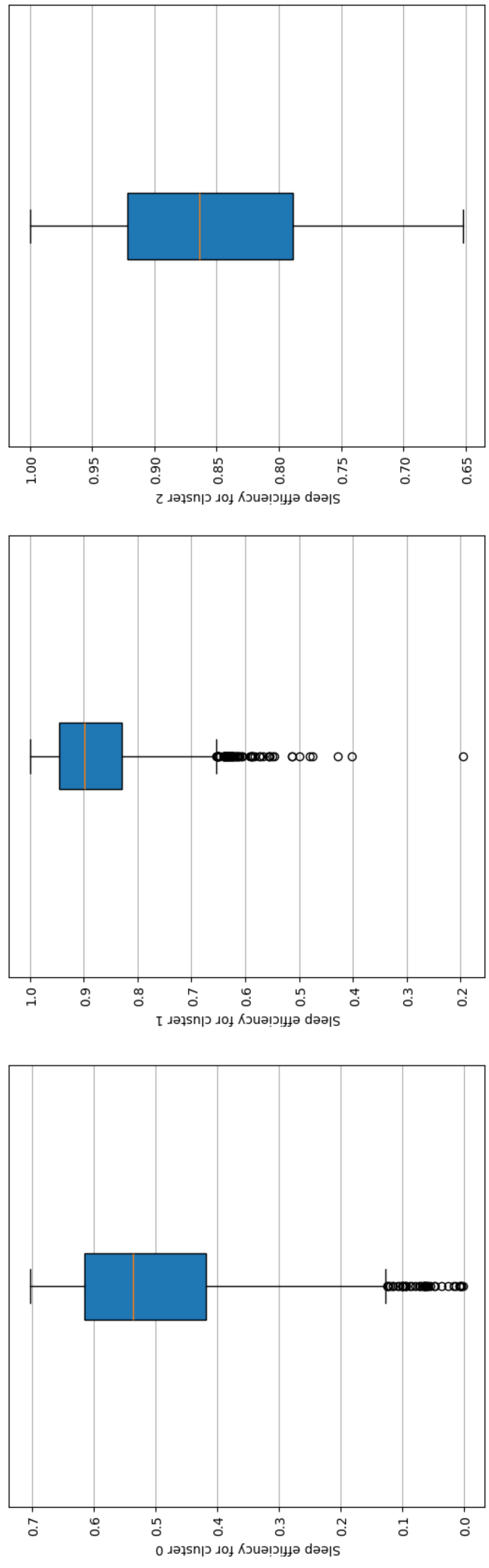

Figure 10: Sleep efficiency for 3 clusters

In figure 10 we can see three different user "profiles". In the first cluster people have a poor sleep efficiency with a median around 0.54 and the lower quartile falling up to 0.13 approximately. This seems to indicate that people in the first cluster is the people having more trouble sleeping, showing a poor sleep efficiency. For the second cluster we have a median sleep efficiency of 0.9, which is good, however a lot of outliers below the lower quartile (around 0.75) are present. Finally, the third cluster shows a median of around 0.87 which is a good sleep efficiency, we have no outliers and the lower quartile is around 0.65 which is low but not terrible.

Of the 362 soundless sessions taken in consideration in the study, 293 are in cluster 1, 55 are in cluster 2 and 14 are in cluster 3. This is a first indicator that people exposed to noise in the ambient is affected severely in the sleep quality.

Lets now get deeper into the data and take a look at how many time the subjects in different clusters are in the different sleep stages:

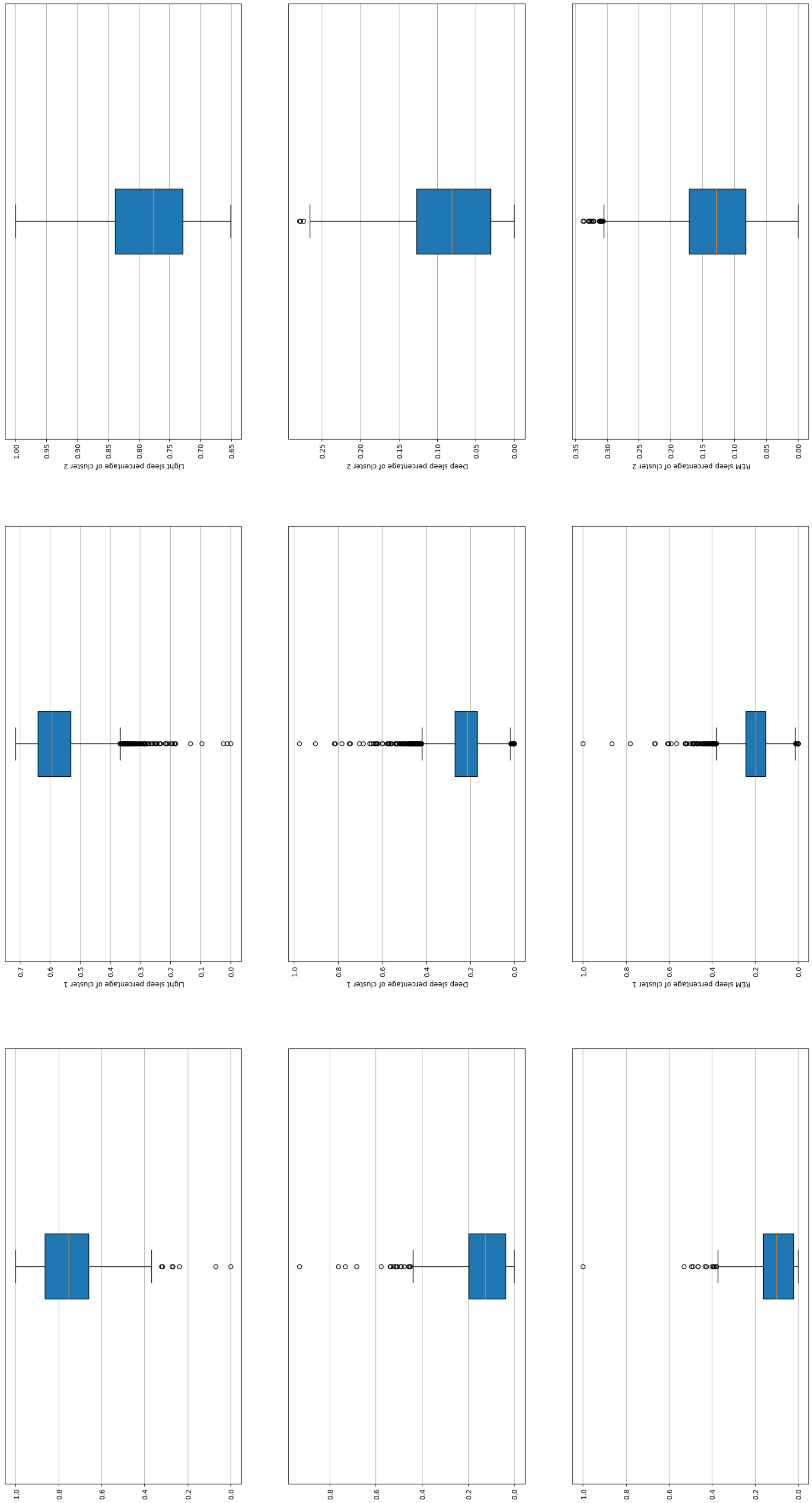

Figure 11: Sleep stages percentages for 3 clusters

The graphic in 11 is more explicative taking into account that light sleep, particularly Stage 2 (N2), makes up approximately 50% of an adult's total sleep time, though this can vary slightly based on individual factors like age and overall health. While Stage 1 (N1) is the very lightest phase where a subject initially drift off, Stage 2 is a more sustained period of light sleep with gradual relaxation and slowed bodily functions. Deep sleep must be around 10% and 25% of the night and REM phase must be in around 20% and 25% of the night.

In the cluster 0 we have the median of light sleep percentage in around 75% which is above the expected percentage, as expected from the graphic 10. Deep sleep for subjects in cluster 1 is around 12% in median, while REM sleep is around 10% what is indicating that subjects that have a worst sleep efficiency have problems too reaching REM sleep phase during the night, either because they are affected by some sleep pathology or are exposed to loud noise in the night.

For cluster 1 we have a median for light sleep time of around a 58%, with a lot of outliers below the the lower quartile, indicating that people in this group is having a better quality in sleep than the previous group, as we can confirm in the graphics for deep sleep percentage (with a median of around 22%) and REM sleep phase which is around 20%.

Finally cluster 2 is kind of a medium group of people, in this group we have people that spend around the 77% of the night in light sleep, only arrive to deep sleep in the 7% of the sleep time and reach REM sleep the 12% of the night. This group is mostly formed of people with some sleep pathology, since only 14 soundless sessions fall into this category.

To confirm our findings we can plot the decibels that the Soundless subjects are exposed to in the different sessions and for the different clusters found:

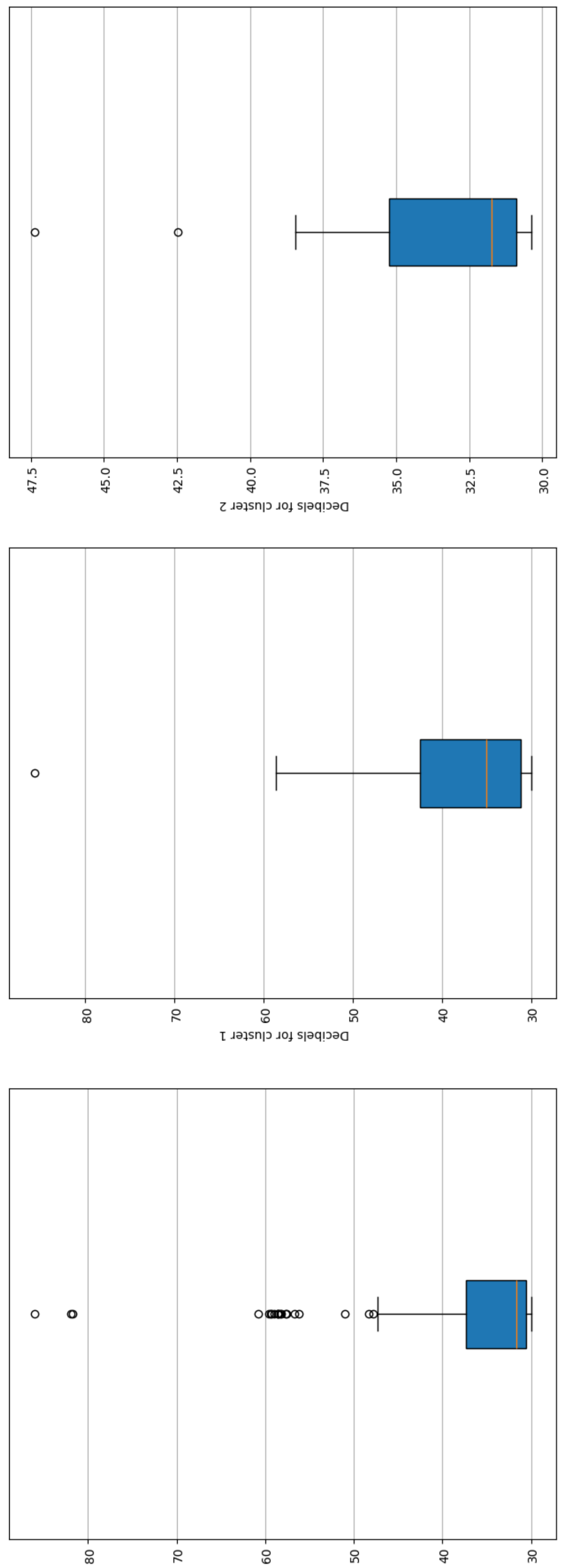

Figure 12: Decibels for soundless users for 3 clusters

In the first cluster, lets remember, the one with more people with trouble on sleep we have a median in decibels of around 32 dB which is a pretty moderated noise, but the upper quartile is above 48 dB and we find a lot of outliers above the upper quartile, 293 soundless sessions fall into this category. The second cluster, this is the one that is able to achieve deep and REM sleep phases a correct percentage of the night, we have a median of around 33 dB an no outliers above the upper quartile (which is around 58 dB), 55 soundless sessions fall into this category. Finally for cluster 2, where we only have 14 Soundless sessions, the decibels are in median around 31 with an upper quartile of around 39 dB.

This seems to confirm that noise in the ambient is not only affecting the sleep efficiency of the subjects, but the sleep quality too. Most of the subjects exposed to loud noise in the night are not able to achieve deep sleep and REM phases a healthy percentage of the night. Particularly in the current study we've found that:

- People exposed to noise can't sleep a healthy percentage of the night.
- People exposed to noise can't reach deep sleep and REM stages a healthy percentage of the night.
- People exposed to noise may be equated to those having an sleep pathology.

Conclusion

Sleep study is a complex topic, and how noise affects sleep increases the complexity of such an study. Some people may be more sensible to noise than another one and sleep quality itself may vary between subjects for many different reasons. Additionally, one subject can be affected by sounds at different frequencies that does not disturb other people. In this document however, we applied a set of techniques that allow us to predict the sleep stage of a subject given an electroencephalogram. This techniques can be adapted or extended to study the noise disturbance on a subject while sleeping. As an instance we could categorize and classify the different arousals found in an electroencephalogram and compare the results with a recording with an standard microphone. The dataset from the Human Sleep Project contains electroencephalograms of subjects with some pathology, it would be nice to have a dataset of healthy subjects to try to expand further our conclusions.

Ideally, with similar montages we can achieve what we want, exposing the subjects to sounds at different frequencies and amplitudes we may get a desirable result for this study. World is not ideal, neither is the data, so with the working dataset of the The Human Sleep project this is a first step towards gaining some knowledge on how the noise affect the sleep quality.

Regarding the soundless dataset, we can say that noise seems to be a crucial factor in sleep quality. People exposed to higher decibels is not able to sleep the most of the night, and people who has been exposed to noise regularly tend to show a lower sleep efficiency than people that does not. Noise is closely related to sleep quality as well, people exposed to noise during the night is not able to reach deep or REM sleep phases a healthy percentage of the time they spend sleeping.

From the 362 Soundless sessions taken in consideration for this study, 293 belong to the worst group studied, the group of people that is not able to sleep most of the time they stay in bed and, when they do, they do not reach deep or REM phases a correct percentage of the time.

We can conclude then that noise seems a determining factor for sleep quality, as presented in this study sounds above 45 dB are enough to disturb a subject enough to not be able to sleep efficiently enough most of the nights, and, we can say too that people exposed to noise regularly may develop an sleep pathology if exposed to noise too much time. Or, in other words, that people exposed to noise while sleeping too many time may show an effectiveness and quality in sleep even lower that people that suffers of an sleep pathology, chronic pain and other sleep disorders.

Future work

As concluded above, the techniques used in this study may be a first step towards gaining some knowledge on how the noise affects the sleep quality. By classifying the sleep stages of a subject given an electroencephalogram we can stablish in which period the noise affects more or less a subject. Besides, the generalization on sleep stage classification, it is, being able to predict the sleep stage of a subject given other subjects data is a great capability, the way we predict sleep stages we can predict arousals given another subject data. Arousal prediction will be another step forward to gain knowledge on how noise affects the sleep quality.

With a bigger dataset we could use soundless data to make a deeper study on the noise in the ambient, probably by taking the power of sound at different frequencies and trying to detect abnormalities in the signals registered by the bracelet, however, an electroencephalogram is ideal to study how sleep occurs during the resting hours.

Further steps from this study may include growing the Soundless dataset, detecting arousals from vital signals and trying to stablish a relationship between soundless data and electroencephalogram data. Moreover, the dataset from The Human Sleep Project provides, in some cases, a pre-sleep form. This form could be used to create a user profile, so if soundless users are willing to respond a form similar to those provided by The Human Sleep Project it will be a good point to start to relate data between the two datasets.

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Bibliography

[1] Soundless, available at: https://soundless.app

[2] The Human Sleep Project, available at: https://bdsp.io/content/hsp/2.0/

[3] Power Spectral Density, available at: https://mne.tools/stable/auto_tutorials/clinical/60_sleep.html#design-a-scikit-learn-transformer-from-a-python-function

[4] Time Series Classification from Scratch with Deep Neural Networks: A Strong Baseline, available at: https://arxiv.org/abs/1611.06455

[5] Time series classification, available at: https://keras.io/examples/timeseries/eeg_signal_classification/

[6] Long short-term memory, Wikipedia, available at: https://en.wikipedia.org/wiki/Long_short-term_memory

[7] Multi Layer Perceptron, Wikipedia, available at: https://en.wikipedia.org/wiki/Multilayer_perceptron

[8] Basner, M., & McGuire, S. (2018). WHO environmental noise guidelines for the European region: A systematic review on environmental noise and effects on sleep. International Journal of Environmental Research and Public Health, 15(3), 519.

[9] Basner, M., Müller, U., & Elmenhorst, E.-M. (2008). Single and combined effects of air, road, and rail traffic noise on sleep and recuperation. Sleep, 31(4), 513523.

[10] Brown, A. L., & van Kamp, I. (2009). Response to noise in urban environments: How important are context and perception? Journal of Environmental Psychology, 29(1), 19.

[11] Dratva, J., Zemp, E., Felber Dietrich, D., Bridevaux, P. O., Rochat, T., Schindler, C., & Gerbase, M. W. (2012). Impact of road traffic noise annoyance on health-related quality of life: Results from a population-based study. Quality of Life Research, 21(6), 10551061.

[12] Dzhambov, A. M., Dimitrova, D. D., & Dimitrakova, E. D. (2017). Road traffic noise and annoyance: A systematic review and meta-analysis. Noise and Health, 19(87), 105119.

[13] Elmenhorst, E.-M., Pennig, S., Rolny, V., Quehl, J., Mueller, U., Maass, H., & Basner, M. (2012). Examining nocturnal railway noise and aircraft noise in the field: Sleep, psychomotor performance, and annoyance. Science of the Total Environment, 424, 4856.

[14] Griefahn, B., Bröde, P., Marks, A., & Basner, M. (2006). Autonomic arousals related to traffic noise during sleep. Sleep, 29(4), 531537.

[15] Halperin, D. (2014). Environmental noise and sleep disturbances: A threat to health? Sleep Science, 7(4), 209212.

[16] Muzet, A. (2007). Environmental noise, sleep and health. Sleep Medicine Reviews, 11(2), 135142.

[17] Pirrera, S., De Valck, E., & Cluydts, R. (2010). Nocturnal road traffic noise: A review on its assessment and consequences on sleep. Biological Psychology, 84(3), 229237.

[18] Riedel, N., Babisch, W., Gurris, N., & Greiser, E. (2022). Noise annoyance modifies the association between nocturnal traffic noise and sleep quality. International Journal of Hygiene and Environmental Health, 239, 113902.

[19] Witte, M., Winkler, A., & Hage, S. R. (2021). Nocturnal noise exposure and cognitive decline in older adults: A longitudinal population-based study. Environmental Research, 196, 110973.

[20] Hrayr P. Attarian & Nidhi S. Undevia. Atlas of Electroencephalography in Sleep Medicine.

[21] Suvaditya M. Electroencephalogram Signal Classification for action identification. Available at: https://keras.io/examples/timeseries/eeg_signal_classification/

[22] Alexandre Gramfort, Stanislas Chambon & Joan Massich. Sleep stage classification from polysomnography (PSG) data. Available at: https://mne.tools/stable/auto_tutorials/clinical/60_sleep.html

[23] Zhiguang Wang, Weizhong Yan & Tim Oates. Time Series Classification from Scratch with Deep Neural Networks: A Strong Baseline. Available at: https://arxiv.org/abs/1611.06455

[24] Jiacheng Hou, Zhenjie Song & Ercan Engin Kuruoglu. BrainNetMLP: An Efficient and Effective Baseline for Functional Brain Network Classification. Available at: https://arxiv.org/pdf/2505.11538